QUANSER
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Mechatronic Actuators Trainer

Fundamental Labs – Brushless DC Motor

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Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

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This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

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CE Compliance CE

This product meets the essential requirements of applicable European Directives as follows:

- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Brushless DC Motor Lab

Why Explore a Brushless DC Motor?

There are two types of commonly used DC motors, this lab will help you explore Brushless DC (BLDC). BLDCs are efficient and more durable than regular DC motors. As described in their name, they don't use brushes or a commutator to deliver current to the coils, instead, they use a rotating permanent magnet and stationary coils. This design reduces wear, which makes them last longer, as well as reducing electrical noise while still allowing precise control of torque and speed. By not having brushes and using electronic switching of current, BLDC motors are able to maintain high torque throughout rotation. As such, these motors are highly power efficient, leading to lower energy losses and less heat production.

BLDC's efficiency and reliability make them ideal for applications that require precision and continuous operation, for example, appliances such as washing machines, air conditioners, vacuums, and hard disc drives. However, these motors are harder to control as the actuation is achieved through the interaction of the permanent magnet rotating based on the direction of the generated magnetic field in the stationary coils.

Background

This lab is part of the Fundamentals labs for the Mechatronic Actuators Trainer. These labs will guide you through four different types of motors, so you can understand their operating principle and get you comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**004_motor_bldc**.
- 4_concept_reviews/Mechatronics/**007_rotary_sensors (rotary encoders section)**.

Getting Started

In this lab, you will use the **Actuators Trainer** to learn about the basics of **brushless DC motors**. The lab will guide you through the usage of hall sensors in the motor and the associated logic needed to drive the motor. You will experiment with this to understand how to drive the motor at different speeds and directions. You will also be comparing your results to values found in the datasheet.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have either the provided Brushless DC motor, or another BLDC motor properly wired to be able to interface with the Actuators Trainer.
- The provided BLDC motor is this one. Confirm your motor is wired properly to the 3-pin connector to run it and the 5-pin encoder cable.

- Make sure the Actuators Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/actuators_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.